HUMBER

Faculty of Applied Sciences & Technology

CPAN 370
Business Intelligence

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Data Warehouse

- Data warehouses are centralized repositories that store structured and organized data. For instance, an e-commerce data warehouse might store customer information, sales transactions, and product details. This centralized approach allows for efficient analysis and reporting.
- Extract, Transform, Load (ETL) processes are integral to this, facilitating the movement of data from source systems to a data warehouse.

ETL Process

Extract: In the extract phase, data is gathered from various source systems. For example, an e-commerce company might extract customer data from its CRM, sales data from ERP systems, and product information from databases.

Transform: Data transformation involves cleaning, standardizing, and restructuring the extracted data. For instance, combining and normalizing data formats from different sources to create a unified structure.

Load: The final step is loading the transformed data into the data warehouse. In our example, the transformed customer data, sales data, and product information would be loaded into the respective tables within the data warehouse.

Extract Phase

Definition: The extract phase in ETL involves retrieving data from various source systems, which can include databases, spreadsheets, logs, and other repositories.

Data Sources:

- Examples of data sources include relational databases (e.g., MySQL, Oracle), flat files (CSV, Excel), NoSQL databases (MongoDB, Cassandra), and application logs.
- Each data source may have its own structure and format, making extraction challenging.

Methods of Extraction:

- **Full Extraction:** Entire datasets are extracted from the source system. It's suitable for small datasets but can be resource-intensive for large volumes.
- **Incremental Extraction:** Only new or modified data since the last extraction is fetched. This is more efficient for large datasets, reducing the load on source systems.

Example: Consider an e-commerce company extracting customer data from its Customer Relationship Management (CRM) system.

Data Source: CRM Database (e.g., Salesforce)

Extraction Method:

- **Incremental Extraction:** Only new customer records and updates to existing records since the last extraction are retrieved.

Transform Phase

The transform phase is where data is processed and modified to meet the standards of the data warehouse. Common transformation operations include:

Cleaning: Removing inconsistencies or errors in data, such as fixing typos or missing values.

Aggregation: Summarizing or aggregating data to a higher level, like calculating total sales by region.

Standardization: Ensuring uniformity in data formats, for instance, converting date formats or standardizing units of measurement.

Example: In our e-commerce scenario, during the transform phase, the ETL process might clean up customer addresses, aggregate daily sales into monthly figures, and standardize product categories.

Load Phase

Definition: The load phase in ETL involves taking the transformed data and loading it into the target data warehouse or destination. This phase ensures that the cleaned and processed data is organized and stored in a way that supports efficient querying and analysis.

Data Warehouse Storage:

- The target data warehouse is typically a relational database optimized for analytical queries.
- Tables within the data warehouse structure store specific types of data, such as customer information, sales transactions, or product details.

Loading Methods:

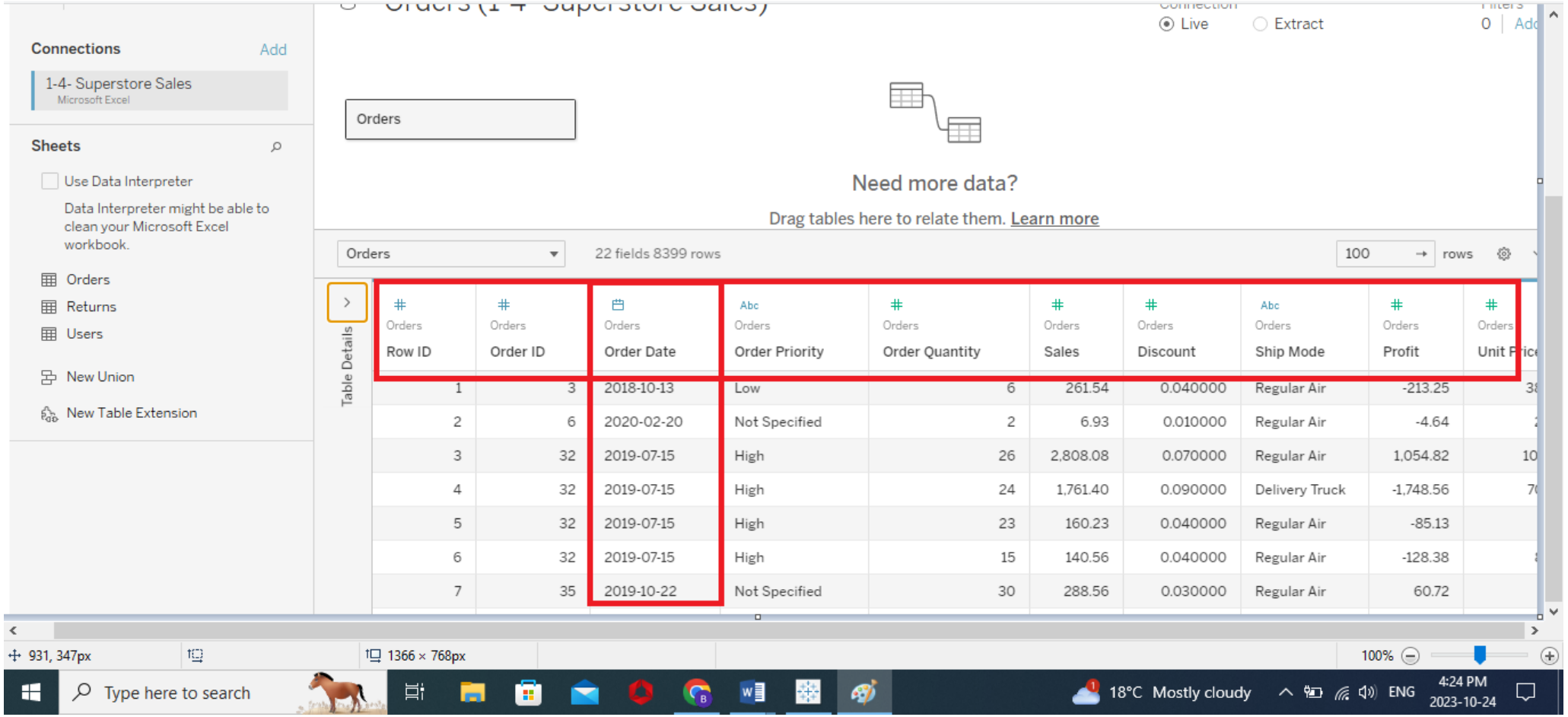
- **Full Load:** All transformed data is loaded into the data warehouse. Suitable for smaller datasets or when a complete refresh is necessary.
- **Incremental Load:** Only new or modified data since the last load is added to the data warehouse, reducing processing time and resource usage.

Example: Continuing with the e-commerce example, after transforming customer data, sales data, and product information in the ETL process, the load phase involves populating corresponding tables in the data warehouse.

Tableau Data Concepts

- Understanding the type and structure of the source data set is essential to efficiently working with the Tableau platform.
- Working with data that you are extremely familiar with lets you visualize it correctly and highlight your analysis.
- Let's start with learning what type of data structure should be used with the tool. Tableau works best with data that is in tables, formatted like a spreadsheet i.e. data that has a vertical format where the field names are located in the header of the columns and the data values are placed in the rows.

Sample-Superstore file



Orders Row ID	Orders Order ID	Orders Order Date	Orders Order Priority	Orders Order Quantity	Orders Sales	Orders Discount	Orders Ship Mode	Orders Profit	Orders Unit Price
1	3	2018-10-13	Low	6	261.54	0.040000	Regular Air	-213.25	38
2	6	2020-02-20	Not Specified	2	6.93	0.010000	Regular Air	-4.64	2
3	32	2019-07-15	High	26	2,808.08	0.070000	Regular Air	1,054.82	10
4	32	2019-07-15	High	24	1,761.40	0.090000	Delivery Truck	-1,748.56	70
5	32	2019-07-15	High	23	160.23	0.040000	Regular Air	-85.13	
6	32	2019-07-15	High	15	140.56	0.040000	Regular Air	-128.38	
7	35	2019-10-22	Not Specified	30	288.56	0.030000	Regular Air	60.72	

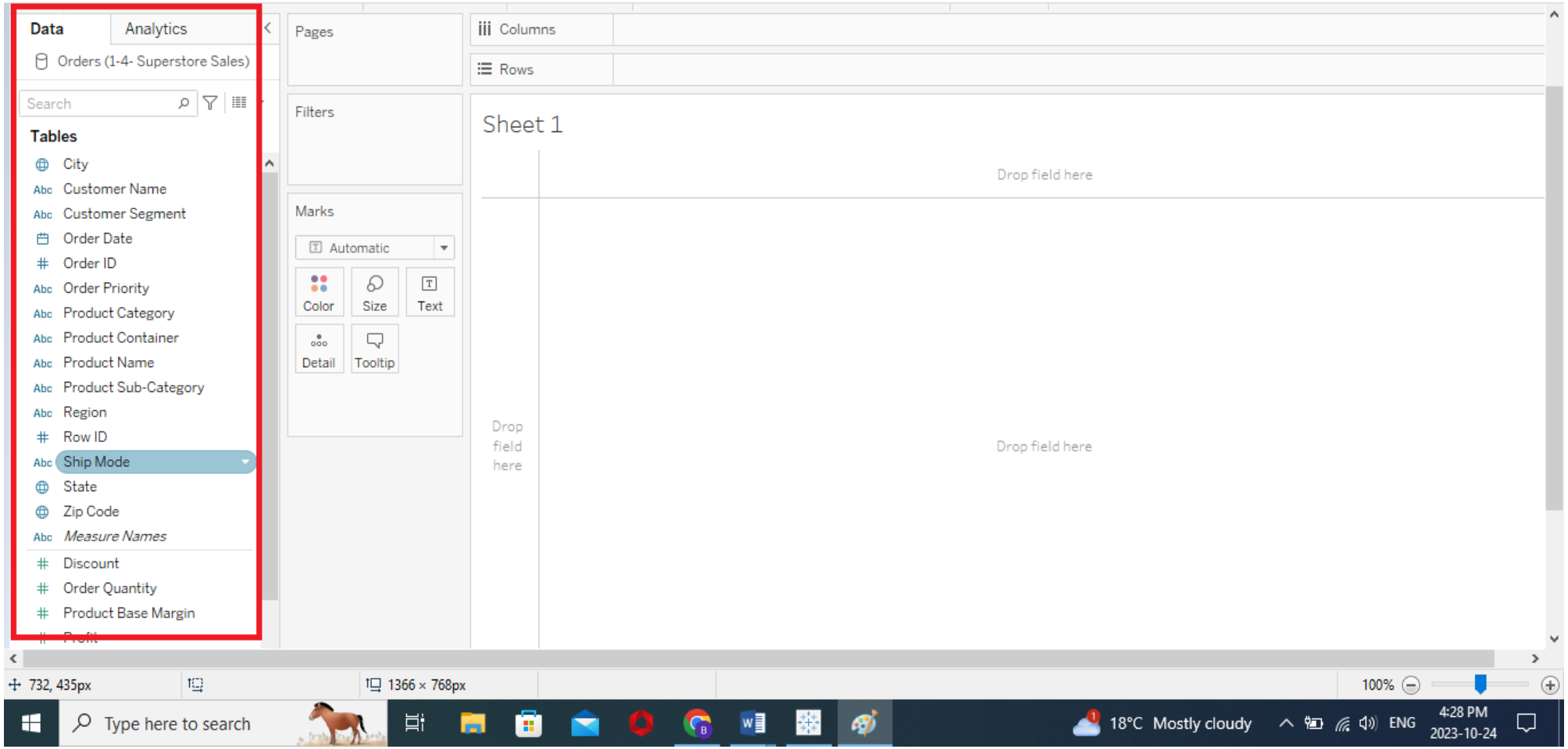
Connected Data Set

Additional things to check when reshaping your data set.

- Make sure that there are no extra headers or footers at top and bottom of the file.
- The first two rows are critical for Tableau to interpret the data source, so it is better to remove the titles and heading inserted above the field names.
- Remove totals in dataset. Tableau can automatically compute totals, so keeping these total columns are unnecessary and may lead to double counting.
- Include date fields in a traditional format rather than ones converted in years, months or other date aggregations. Tableau can automatically aggregate dates so it is more efficient to use the traditional format in order to unlock the full functionality of date fields.

How Tableau reads and categorizes the fields. Upon connecting Tableau to a new data source, a symbol can be seen on top of each field name. These represent the data type of each column. Data types classify the data in a given field and provide information about how the data should be formatted, interpreted and what operations can be performed.

Data Pane

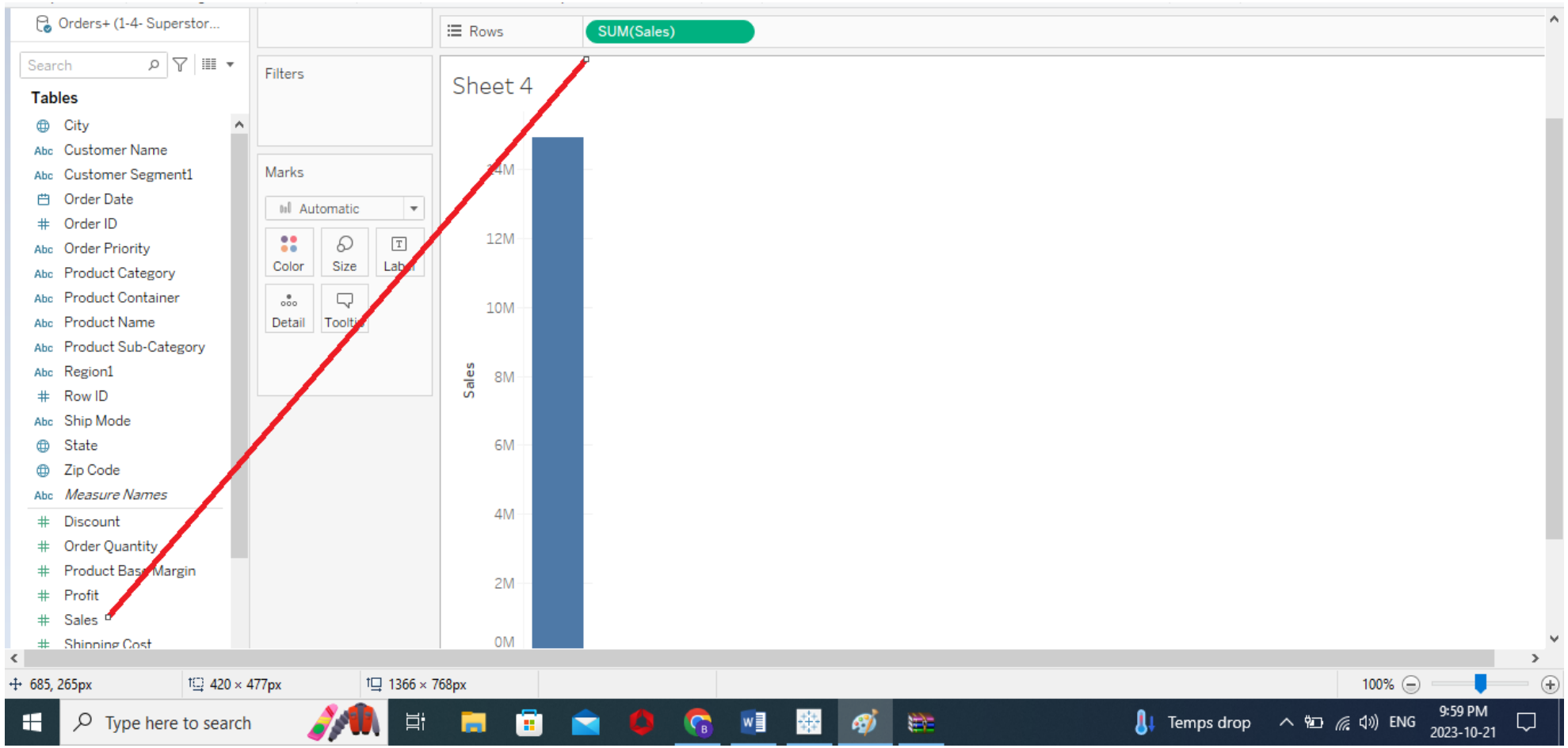

Data
Analytics
Orders (1-4- Superstore Sales)
Search
Tables
City
Customer Name
Customer Segment
Order Date
Order ID
Order Priority
Product Category
Product Container
Product Name
Product Sub-Category
Region
Row ID
Ship Mode
State
Zip Code
Measure Names
Discount
Order Quantity
Product Base Margin
Pages
Filters
Marks
Automatic
Color
Size
Text
Detail
Tooltip
Columns
Rows
Sheet 1
Drop field here
Drop field here
Drop field here
732, 435px
1366 × 768px
100%
Type here to search
18°C Mostly cloudy
ENG
4:28 PM
2023-10-24

Measures and Dimensions

The data fields are categorized by Tableau into roles of either Dimensions or Measures.

- Dimensions have a blue data type icon and a blue pill which is visible when you hover over the field name. Measures have a green data type icon and a green pill.
- Dimensions contain qualitative values which can't be measured, but are instead described. Dimensions are often things like city or country, eye color, category, team name. You can use dimensions to categorize, segment and reveal the details in your data. Also, dimensions affect the level of detail in the view.
- Measures contain numeric quantitative values that you can measure and record using numbers. Measures can be aggregated.
- In the data pane, dimensions are listed first by default followed by the measures which are separated by a line.

Measures


Orders+ (1-4- Superstor...
Search
Tables
City
Customer Name
Customer Segment1
Order Date
Order ID
Order Priority
Product Category
Product Container
Product Name
Product Sub-Category
Region1
Row ID
Ship Mode
State
Zip Code
Measure Names
Discount
Order Quantity
Product Base Margin
Profit
Sales
Shipping Cost
Filters
Marks
Automatic
Color
Size
Detail
Rows
SUM(Sales)
Sheet 4
14M
12M
10M
8M
6M
4M
2M
0M
Sales
685, 265px
420 × 477px
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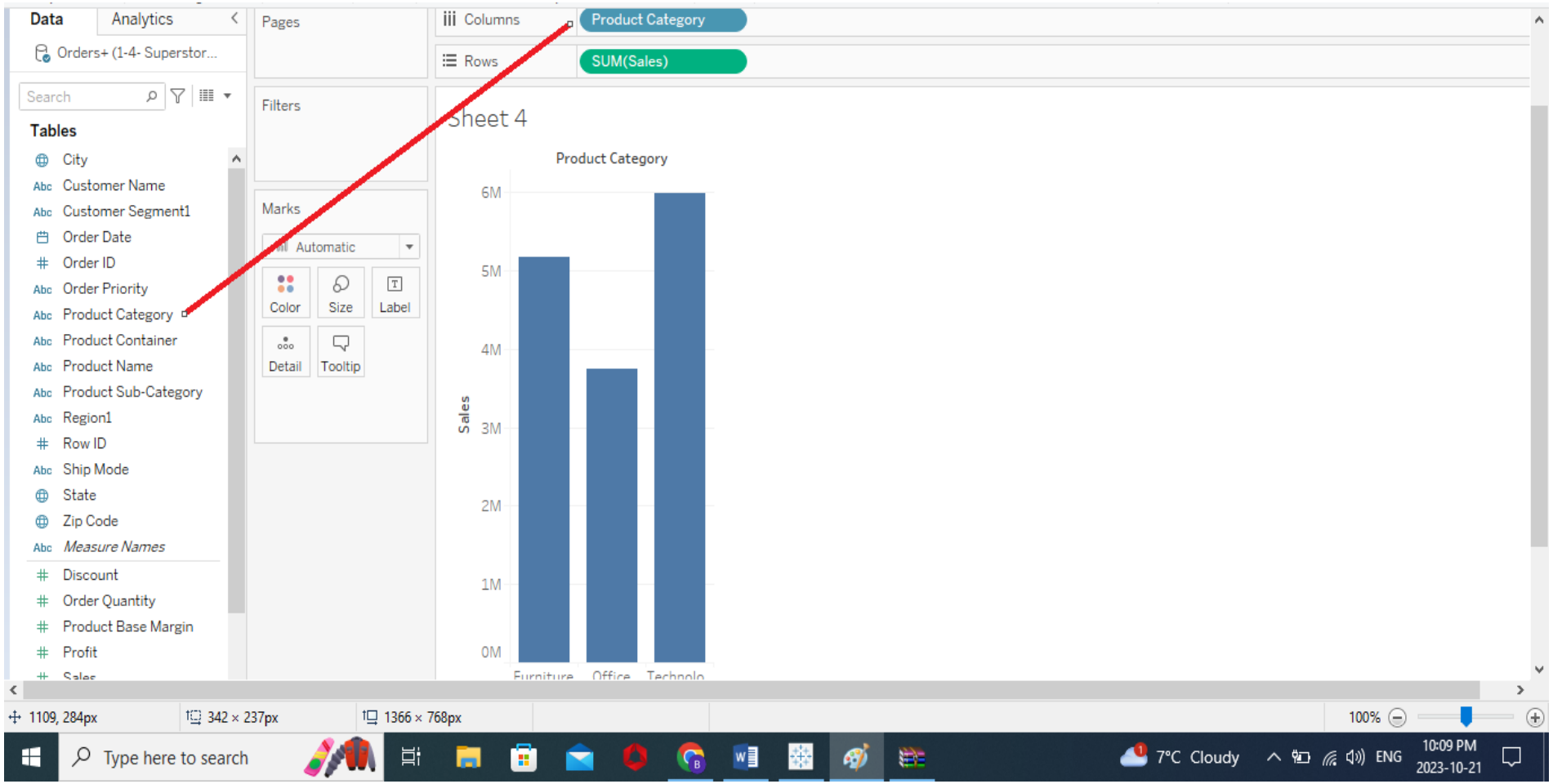
2023-10-21

Dimensions


Data
Analytics
Orders+ (1-4- Superstor...
Search
Tables
City
Customer Name
Customer Segment1
Order Date
Order ID
Order Priority
Product Category
Product Container
Product Name
Product Sub-Category
Region1
Row ID
Ship Mode
State
Zip Code
Measure Names
Discount
Order Quantity
Product Base Margin
Profit
Pages
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Columns
Product Category
Rows
SUM(Sales)
Sheet 4
Product Category
6M
5M
4M
3M
2M
1M
0M
Sales
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2023-10-21

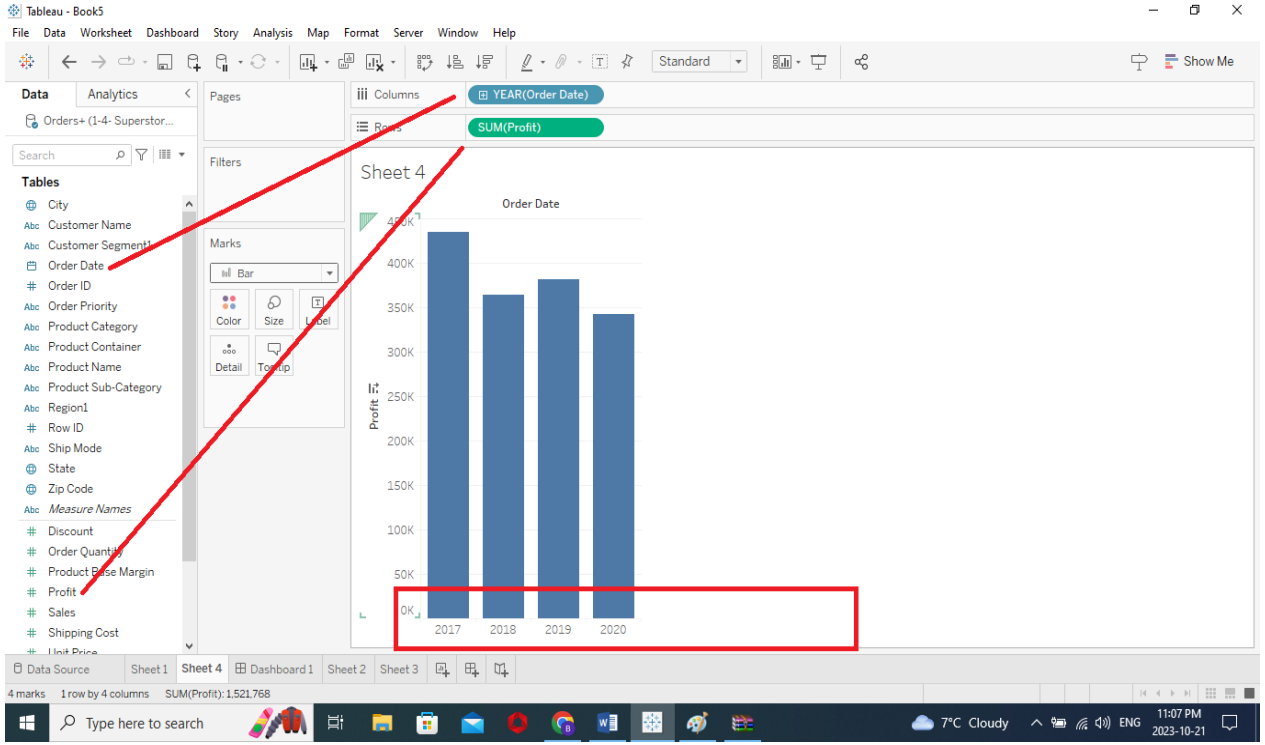
Discrete and Continuous

Tableau represents the data differently depending on whether the Dimension and Measure are discrete or continuous.

- Discrete values are individually separate and distinct, such that it can only contain certain values which are countable. For example, different types of products are discrete, sales of each product is also considered as discrete.
- Continuous values can be infinite; it can take any value within a specific range, time or interval. A person's age is considered continuous, the time in a race and temperature of a freezer are also continuous values.

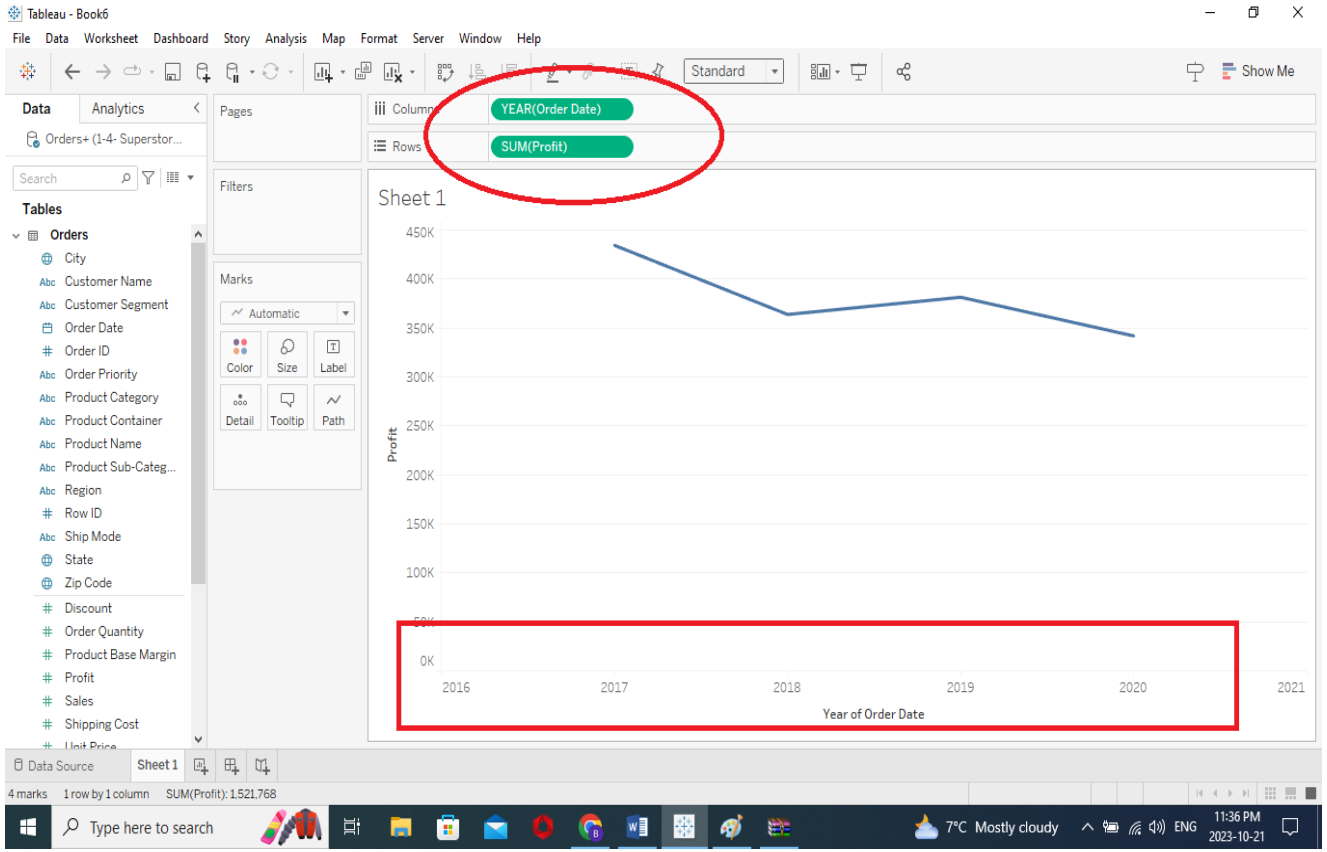
Discrete and Continuous

In this bar chart we have a continuous measure of profit and a discrete dimension of year of order date. As you can see the graph has no x-axis line since the year of order date is set as discrete or distinct year value. The year is treated as a header or label for each bar.



Discrete and Continuous

Now let's compare it to this second chart which uses the same data. In this line chart, we still use a continuous measure of sum of profit but the year of order date is also set as continuous as implied by its green pill.



Connect to Data Source

A data source is essentially the sum of your data, the connection information, the names of tables or sheets containing the data and the customizations that you will make on top of data to work with in Tableau.

A Tableau Data Source may contain multiple data connections to different databases or files. Connection information includes where the data is located such as a file name and a path or a network location and details on how to connect to your data such as database server and server sign-in information.

Things to consider

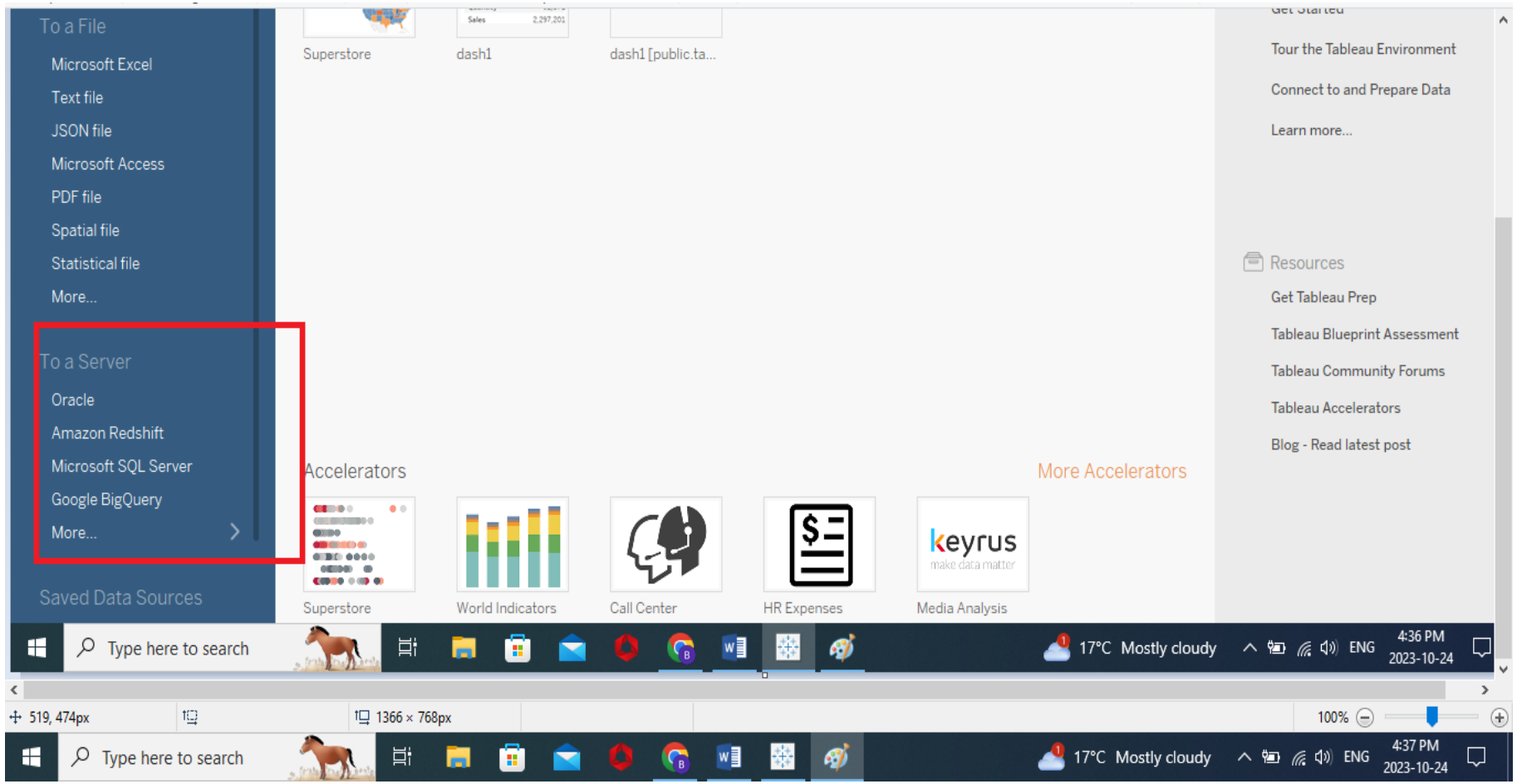
There are several things to consider before connecting to a Tableau Data Source.

- **Location and access;**
- **Shape and cleanliness;**
- **Data model and combining the data;**
- **Performance and freshness;**

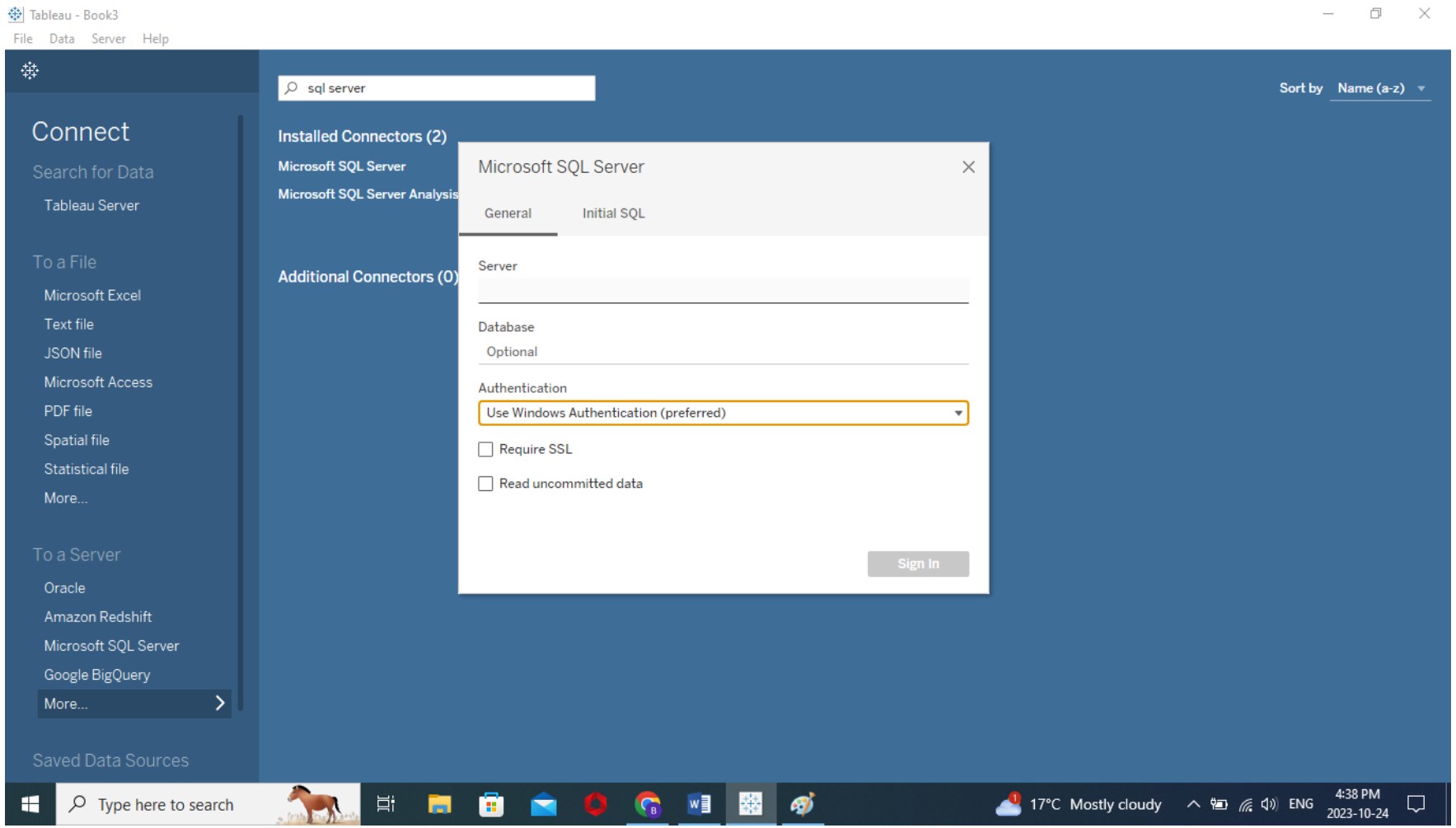
Data Source Support

- Tableau supports many types of data stored in different places, but the catch is this will depend on the type of Tableau product that you have installed.
- Tableau desktop professional has the complete data source connections which includes a native connection to server hosted data sources. It has several options of relational multi-dimensional and cloud-based databases and repositories.
- Tableau desktop personal and public on the other hand can only connect to flat files, Google Sheets and web data connectors.
- Databases are great to use as they already have a defined structure where data types are already defined.
- At times flat files are provided as data source. Flat files can be prone to wrong data formats and they are sensitive to change in column names.

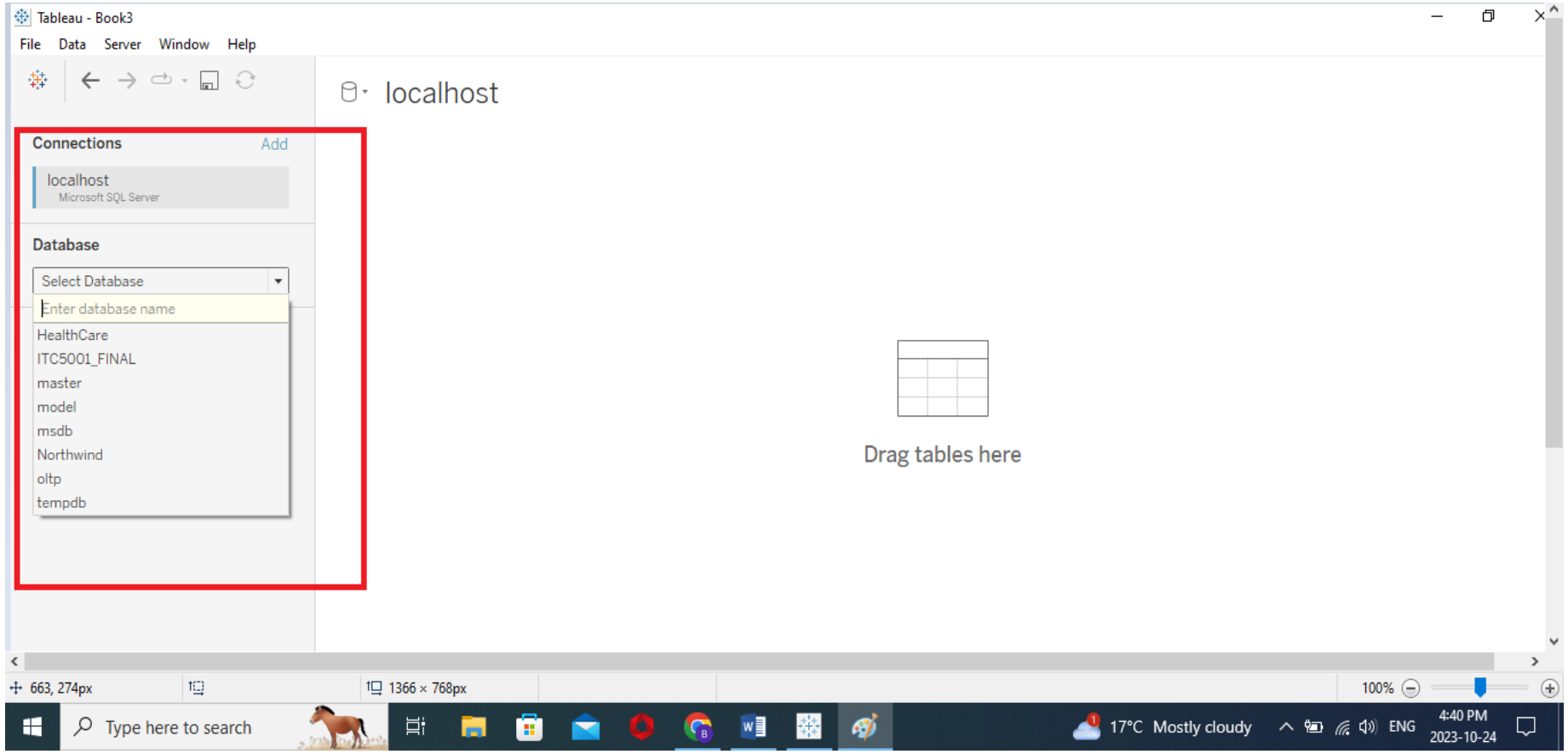
To connect Tableau to a database



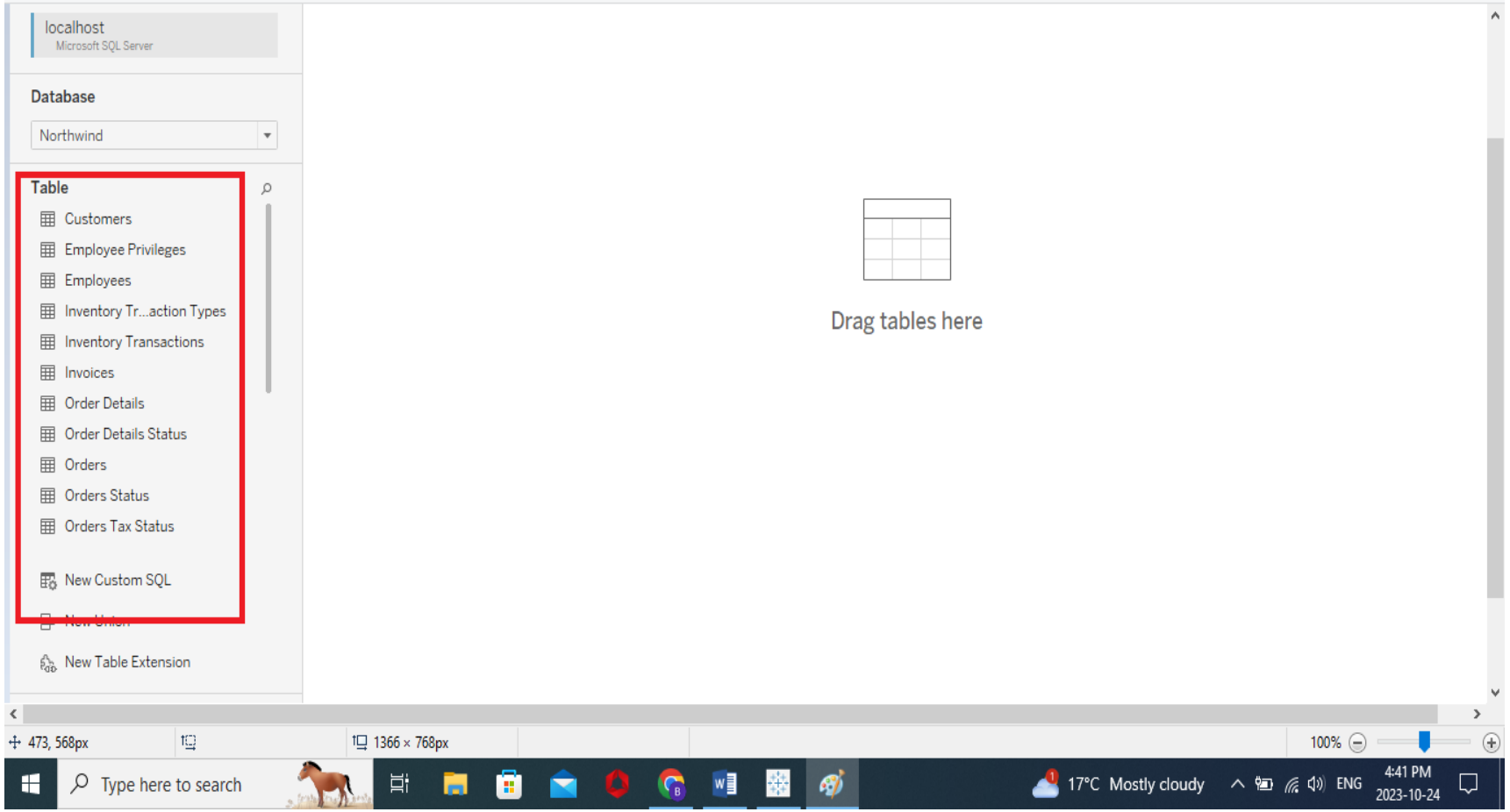
Connection Information



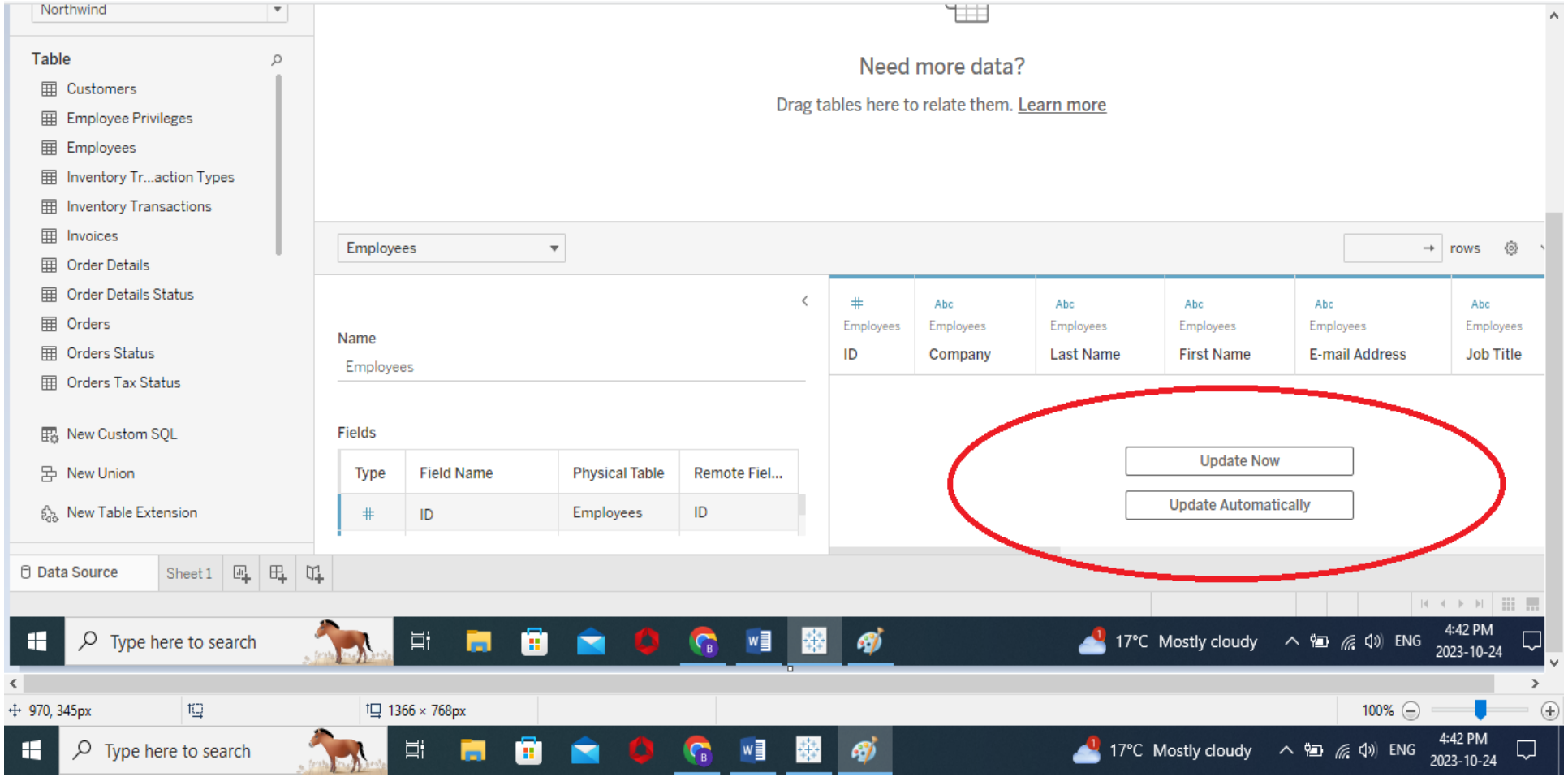
Database Name



Select Tables



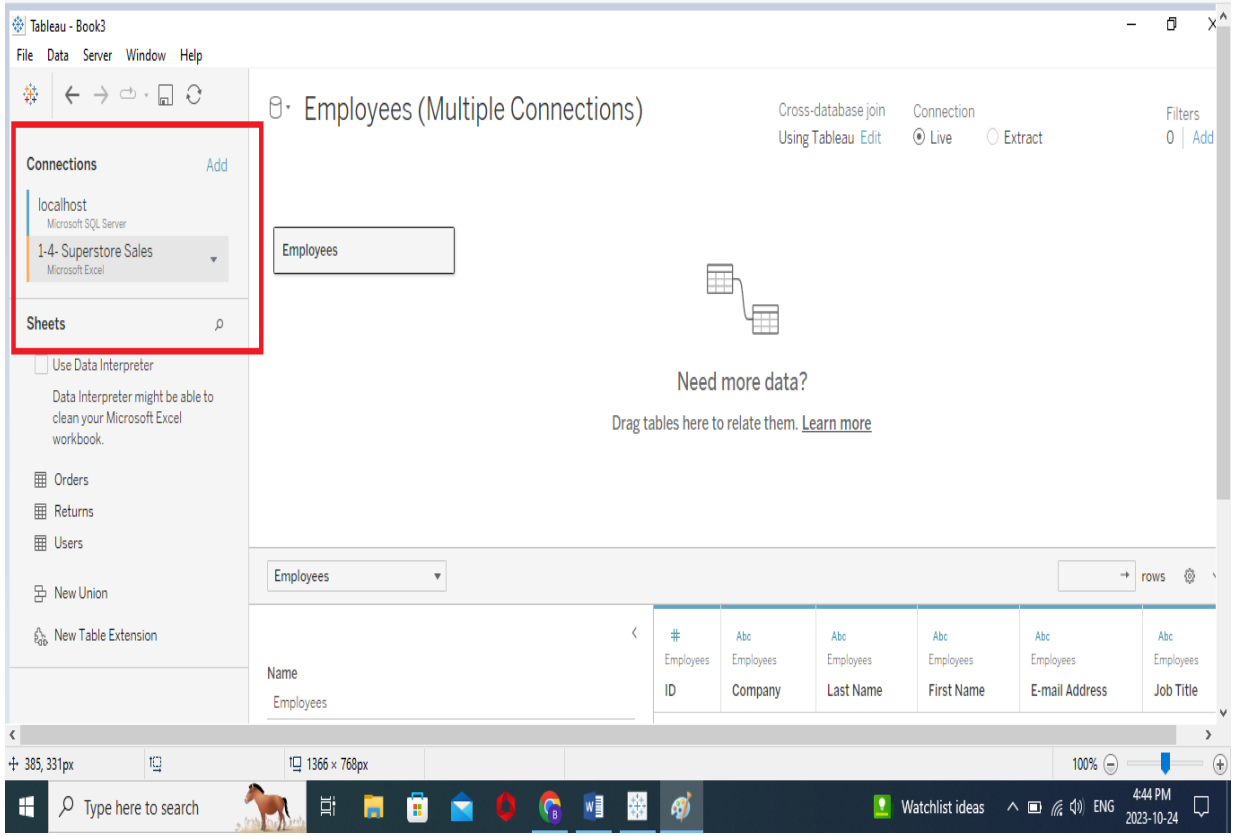
Update Now


Northwind
Table
Customers
Employee Privileges
Employees
Inventory Tr...action Types
Inventory Transactions
Invoices
Order Details
Order Details Status
Orders
Orders Status
Orders Tax Status
New Custom SQL
New Union
New Table Extension
Need more data?
Drag tables here to relate them. Learn more
Employees
rows
Name
Employees
Fields
Type
Field Name
Physical Table
Remote Fiel...
ID
Employees
ID
Employees
ID
Employees
Company
Employees
Last Name
Employees
First Name
Employees
E-mail Address
Employees
Job Title
Update Now
Update Automatically
Data Source
Sheet 1
Type here to search
17°C Mostly cloudy
ENG
4:42 PM
2023-10-24
970, 345px
1366 × 768px
100%

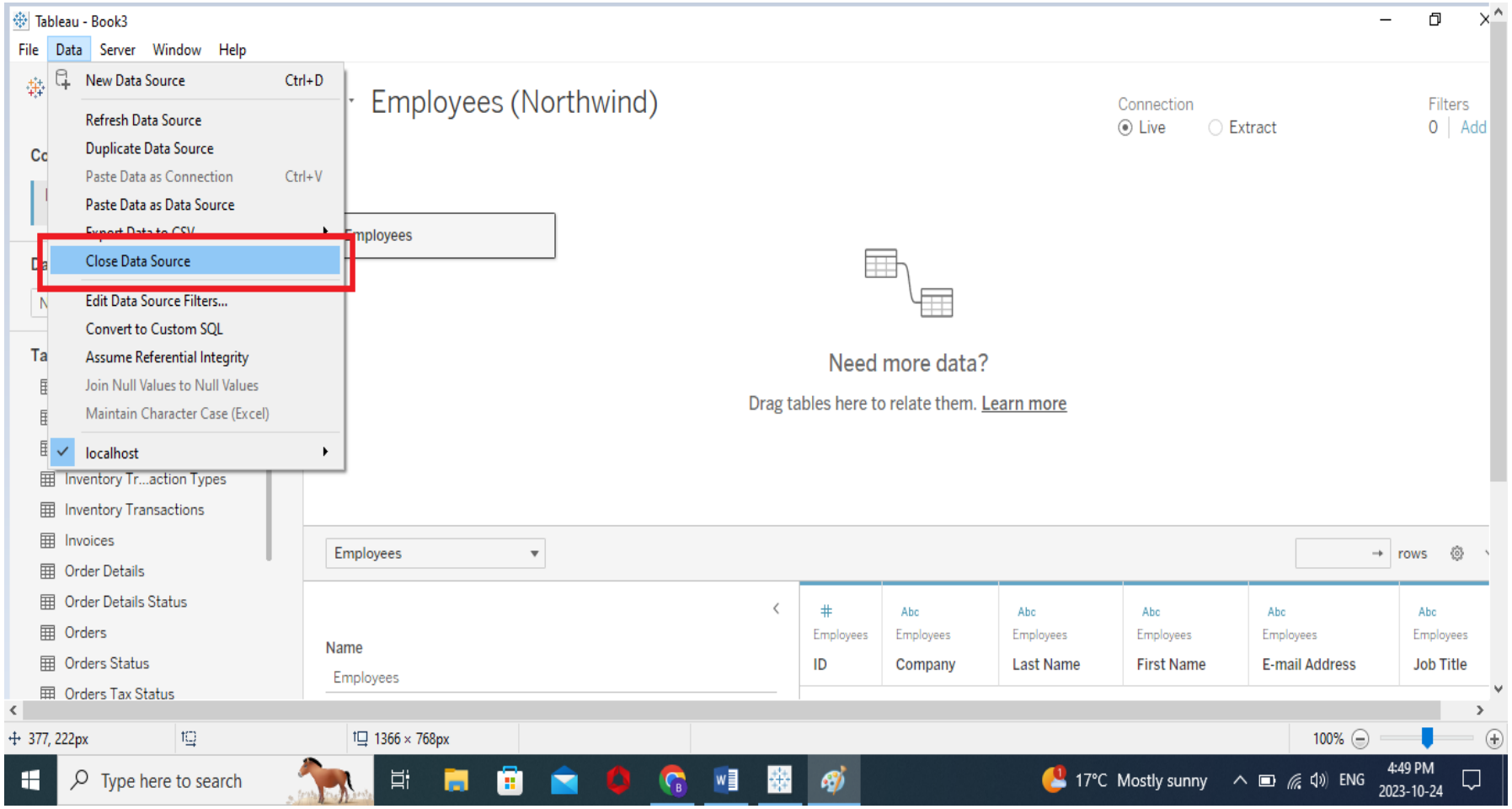
Data Sources in Tableau

Look at the connections section on the upper left. The database section lists the current files or databases that you have connected to Tableau.

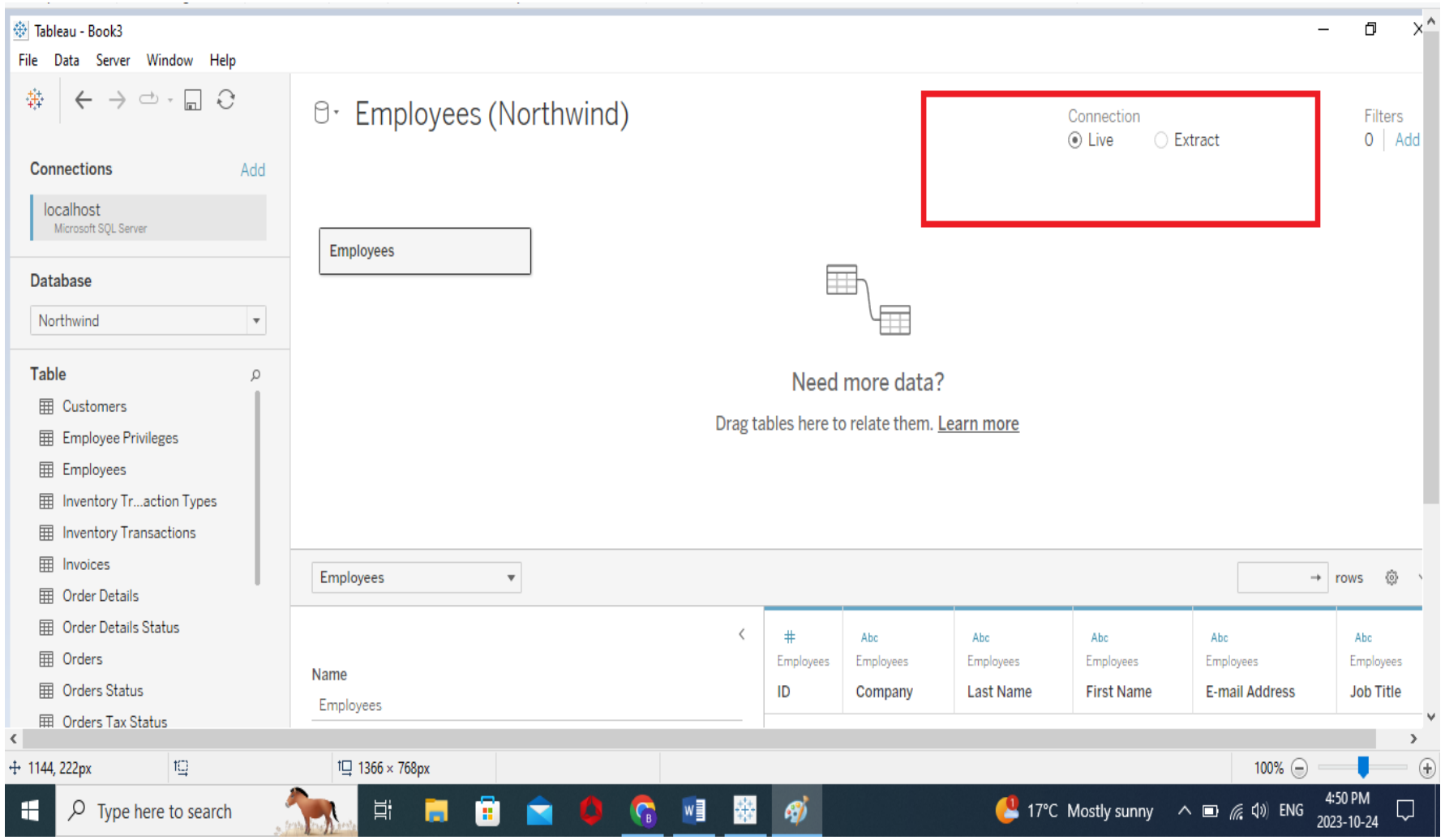
The first data source that you add will be tagged with the blue color. Notice that if we add a second data connection to it, the new connection will have a different coloured tag. This is done in order to easily identify and differentiate each individual connection.



Remove a Data Source



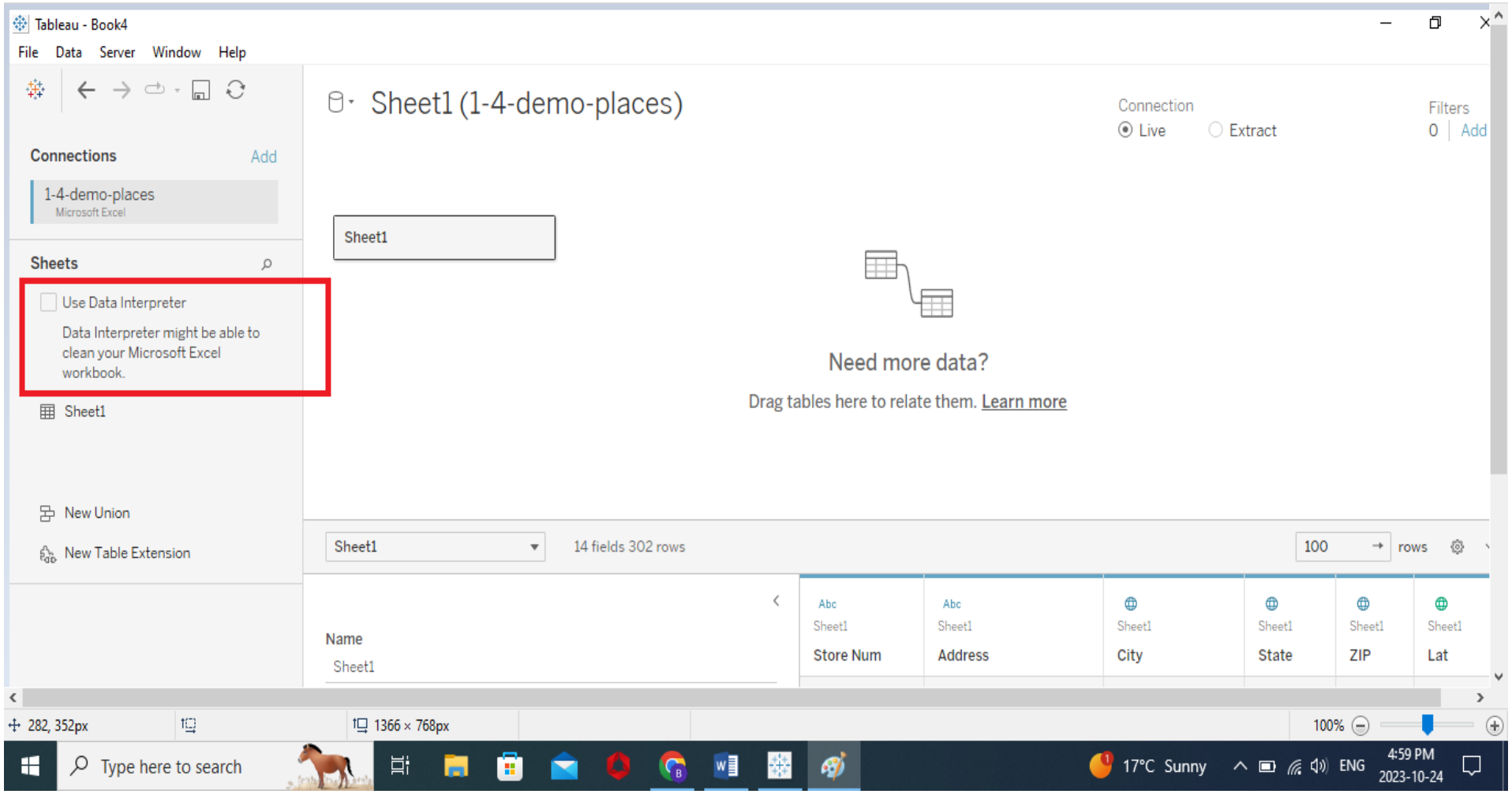
Live and Extract Connections



Live and Extract Connections

- Live connections offer the convenience of real-time updates where any changes in the data source automatically reflect in Tableau, but Live connections also rely on the database for all queries. Thus, it is not always optimized for fast performance.
- Your data queries will only be as fast as the database itself. The workbook speed can also be affected by your network speed, traffic and any custom SQL.
- Complex workbooks with Live connections can also stress some traditional databases.
- Extracts on the other hand are a snapshot of data optimized for aggregation and loaded into system memory to be quickly recalled for visualization. When you use the Extract connection type, Tableau creates a TDE or Tableau Data extract file.

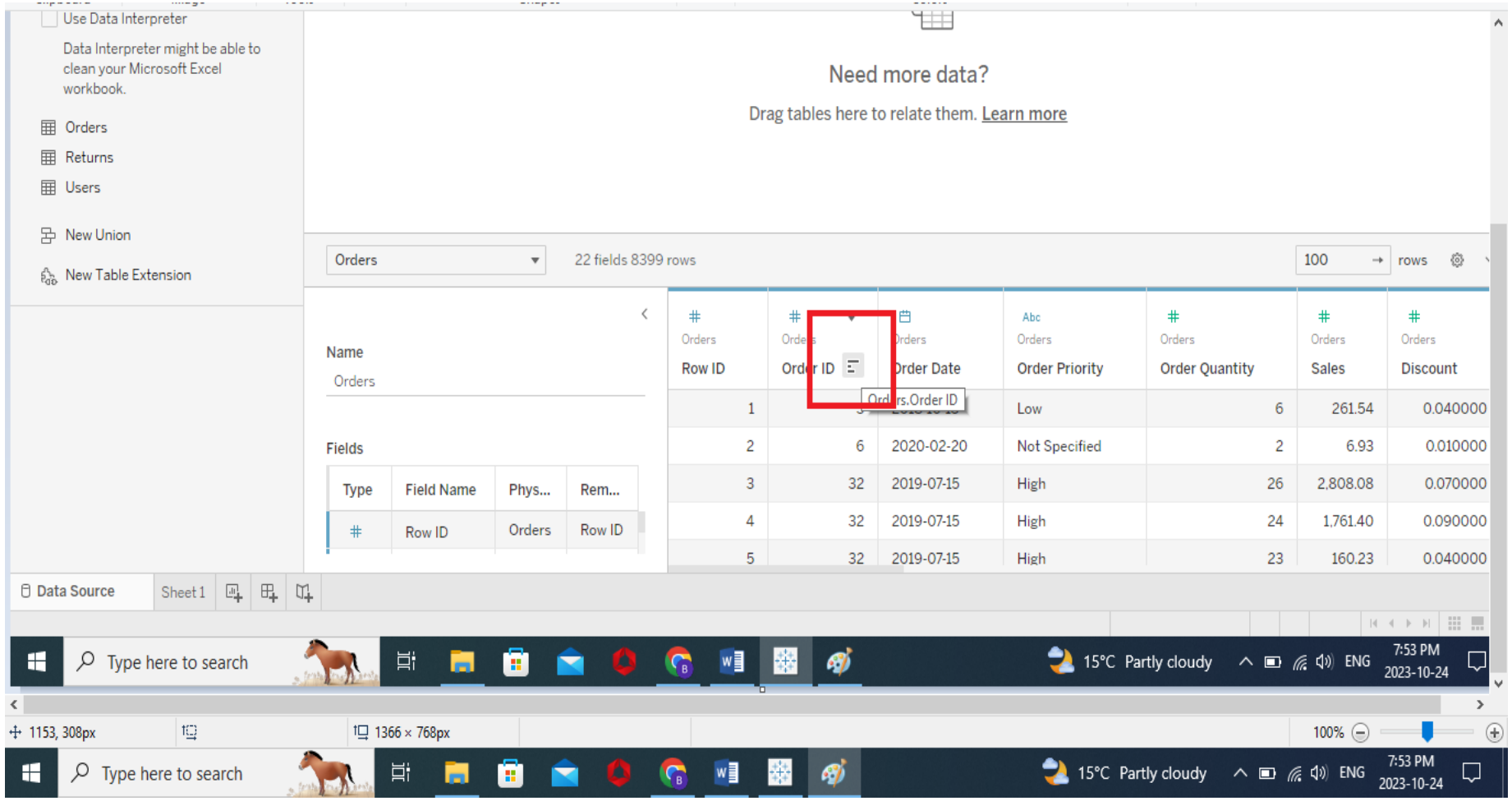
Data Interpreter



Data Interpreter

- The Data Interpreter helps you clean data by detecting titles, footnotes, empty cells and other extra columns/rows to identify the actual fields and values in your data set.
- To check all of the changes done by the Data Interpreter, you can click on the “review the results” which will output an Excel file that lists the changes using the original copy of data with color-coded cells.
- If the modifications done by the Data Interpreter are not enough, you can always apply basic data formatting using the functions in the data pane.
- To rename a column, click its drop down button, select the rename function then type in the new column name. You can also rename a field by simply double-clicking its name and typing in a new field name.
- If you wish to bring back the original file names, click the drop down for the column and select reset name.
- Apart from the rename function, you can also apply a sort. Hover over a specific field's name until the sort button appears on its right.

Data Interpreter



Thank You